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Experimental secure multiparty computation from quantum oblivious transfer with bit commitment

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Secure multiparty computation enables collaborative computations across multiple users while preserving individual privacy, which has a wide range of applications in finance, machine learning and healthcare. Secure multiparty computation can be realized using oblivious transfer as a primitive function. In this paper, we present an experimental implementation of a quantum-secure quantum oblivious transfer (QOT) protocol using an adapted quantum key distribution system combined with a bit commitment scheme, surpassing previous approaches only secure in the noisy storage model. We demonstrate the first practical application of the QOT protocol by solving the private set intersection, a prime example of secure multiparty computation, where two parties aim to find common elements in their datasets without revealing any other information. In our experiments, two banks can identify common suspicious accounts without disclosing any other data. This not only proves the experimental functionality of QOT, but also showcases its real-world commercial applications.

Secure multiparty computation (MPC) in cryptography[1–3] enables multiple parties to jointly compute a specific function on their respective data without revealing any additional information about their private data. MPC plays a crucial role in fintech. For example, in fraud detection and anti-money laundering, financial institutions can use MPC to jointly compute the intersection of data sets, such as blacklists and suspicious transactions, while keeping each party's private information confidential. Beyond fintech, MPC also finds applications in privacy-preserving machine learning[4], where organizations collaborate on training models using sensitive data without exposing individual datasets. Similarly, in the secure computation of genetic data[5], multiple parties can perform analyses on shared genetic datasets while maintaining the privacy of participants' personal information. These use cases demonstrate MPC's versatility in facilitating secure, privacy-conscious computations across various industries.

Secure MPC can be realized using oblivious transfer (OT) as an underlying protocol[6]. OT allows a sender to transmits data such that the receiver learns only the selected piece while the sender remains unaware of the choice. 1-out-of-2 OT, the simplest form, is illustrated in Fig. 1. The security of a classical OT protocol is based on the conjectured computational hardness assumptions[7,8], such as the RSA problem and discrete logarithm, which are vulnerable to quantum attacks using Shor's algorithm. Inspired by quantum cryptography, especially quantum key distribution (QKD), it is natural to explore a quantum analog to OT, which we refer to as quantum OT (QOT)[9].

While unconditional security is unattainable with quantum alone[10,11], QOT protocols generally depend on weaker cryptographic assumptions for their security proofs compared to their classical counterparts. After QOT protocol[12] was proposed, the QOT protocol in the noisy storage model was extensively studied[13–15] and its experimental implementation was achieved in ref. 16. However, this protocol becomes vulnerable when large and reliable quantum memory becomes available[17]. Rather than restricting the adversary's technological capability, QOT can rely on the existence of cryptographic functionality primitives such as bit commitment[18] to achieve security against quantum attacks. In addition, although there were several experimental demonstrations of QOT[16,19–21], to our best knowledge, no implementations of secure MPC based on QOT for addressing real-world problems have been demonstrated.

In this paper, we experimentally implement a QOT protocol from[22–24], using a bit commitment scheme to ensure its theoretical security against quantum attacks. This experimental implementation is adapted based on decoy-state QKD protocol. We then leverage our QOT protocol for multiparty secure computations. We demonstrate its application to the private set intersection (PSI)[25,26] problem, allowing two banks to securely determine the intersection of the blacklist of one bank and the new client list of the other, without revealing any additional information.

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Results

QOT Protocol

In this paper, we experimentally implement a 1-out-of-2 QOT protocol and then apply it to solve one of the most practically motivated MPC problems – the PSI problem. Different from previous QOT protocols[13–16,27], we do not assume the adversary has access only to either noisy or bounded quantum memory. Instead, our protocol is secure against any classical or quantum attacks with polynomial time complexity. The procedure of this protocol is summarized in Fig. 2 and the detailed procedure adopting decoy-state BB84 protocol is presented in Methods 4.2. The key difference between our QOT protocol and the one in the noisy storage model is that Bob must commit all his measurement bases through a bit commitment scheme.

Here, we analyze the correctness and security of our QOT protocol, taking into account the practical limitations of quantum devices. The correctness of the protocol refers to the fact that both honest parties can successfully accomplish this protocol with high probability when employing the following experimental setup in Sec.2.2. One essential imperfection introduced by Alice's quantum device is the single photon source approximated by a weak coherent state, leading to photon missing at Bob's side in certain rounds. To make the ratio of Bob's photon missing for each decoy state source fall within Alice's acceptable range, Alice should choose the range as $[p_0^B - \epsilon, p_0^B + \epsilon]$, where $p_0^B = p_{\geq 1}^A - \eta - p_{dark}$ is the expected ratio of photon clicks at Bob's detector, $p_{\geq 1}^A$ denotes the probability of generating at least one photon at Alice's decoy state source, η denotes the transmission loss and p_{dark} denotes the dark count rate of Bob's detector. To make the test pass with probability at least $1-\delta$, Alice should choose $\epsilon \geq \sqrt{1/N \log(1/\delta)}$.



Fig. 1 | 1-out-of-2 Oblivious Transfer. Alice has two possible messages, m_0 and m_1, to send to Bob. Bob chooses a bit c from the set containing zero and one, and at the end of the process, Bob receives message m_c, without Alice learning the value of c.

Another important source of error in the protocol is bit flip errors induced by transmission noise and measurement errors. Bit flip errors affect two steps in the protocol: they may cause a false failure to pass Alice's test in steps 3-5 with probability p_{fpass}, and if bit flip errors occur in more than t bits, they lead to failure of error correction in step 9, with probability p_{fcorrect}. The total failure probability is bounded above by $p_{\text{fpass}} + p_{\text{fcorrect}}$.

The security of the protocol ensures that a dishonest Alice learns no information about Bob's chosen bit c, while a dishonest Bob, when dealing with honest Alice, gains no knowledge of the message $\boldsymbol{m}_{1-c}$, which he did not select. Our OT protocol guarantees unconditional security against a malicious Alice. In other words, Alice does not get any information about Bob's chosen bit c.

For a malicious Bob, if an imperfect single-photon source emits two or more photons with nonzero probability, he could receive multiple photons during a transmission. In this case, Bob could measure one photon immediately and store the others in his quantum memory, allowing him to measure them after Alice announces her measurement bases $\boldsymbol{\theta}$. Consequently, Bob can correctly deduce the value of x_i for both $i \in I_0$ and $i \in I_1$, thus obtaining information about $\boldsymbol{m}_{1-c}$, which he was not supposed to learn. To exploit this vulnerability, a malicious Bob could underreport the number of single-photon communications, discarding cases where he received only one photon without notifying Alice. This strategy increases the proportion of multiple-photon events compared to the total effective rounds. Such an attack is analogous to the photon number splitting (PNS) attack used by eavesdroppers in QKD[28,29]. However, Bob's PNS attack would modify the probability of photon clicks at his detector. For each decoy state, because Bob is unaware of Alice's probability $p_{\geq 1}^A$, his attack introduces deviations in the photon click statistics that he reports. These deviations would fall outside of Alice's acceptable range, $[p_0^B - \epsilon, p_0^B + \epsilon]$, and can be used by Alice to detect Bob's PNS attack.

We now analyze the probability that a malicious Bob can circumvent the bit commitment scheme and get both of Alice's messages. Bob might attempt to delay his measurement in step 2 until Alice announces the correct basis in step 6. However, he must first bypass the verifications in steps 3–5. Assuming a scenario most favorable for Bob, where the error rate $p_e = 0$, we

Protocol QOT

1. *Alice* randomly selects $\boldsymbol{x} \in \{0,1\}^n$ and $\boldsymbol{\theta} \in \{0,1\}^n$, and prepares and sends polarization-encoded single-photon state $|\boldsymbol{x}\rangle_{\boldsymbol{\theta}}$.
2. *Bob* randomly selects $\tilde{\boldsymbol{\theta}} \in \{0,1\}^n$ and measures $|\boldsymbol{x}\rangle_{\boldsymbol{\theta}}$ in the basis of $\tilde{\boldsymbol{\theta}}$, obtaining $\tilde{\boldsymbol{x}} \in \{0,1\}^n$, where $|0\rangle_0 := |0\rangle$, $|1\rangle_0 := |1\rangle$, $|0\rangle_1 := |+\rangle$, and $|1\rangle_1 := |-\rangle$.
3. *Bob* commits to all $\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{x}}$ by sending $c_i = h(\tilde{\theta}_i, \tilde{x}_i, \boldsymbol{r}_i)$ for $i \in \{1, \ldots, n\}$, where h is a hash function and each $\boldsymbol{r}_i$ is a random bit string.
4. *Alice* randomly selects a subset a subset $T \subset [n] := \{1, 2, \ldots, n\}$ of size $|T| = \alpha n$, where $0 < \alpha < 1$. *Bob* opens c_i for $i \in T$, and *Alice* verifies if for $i \in T$, $x_i = \tilde{x}_i$ when $\theta_i = \tilde{\theta}_i$ holds true with a ratio of at least $\beta > 1/2$. If the test is passed, then they continue; otherwise, Alice aborts the protocol.
5. Both parties discard the test sets.
6. *Alice* sends $\boldsymbol{\theta}$ to *Bob*.
7. *Bob* chooses a choice bit $c \in \{0,1\}$, divides the set $[n] \setminus T$ into good and bad ones, i.e., $I_c = \{i \mid \theta_i = \tilde{\theta}_i\}$ and $I_{1-c} = \{i \mid \theta_i \neq \tilde{\theta}_i\}$, and sends I_0, I_1 to *Alice*. The bit strings $\boldsymbol{x}$ is then divided into two $\boldsymbol{x}_{I_c}$ and $\boldsymbol{x}_{I_{1-c}}$ on I_c and I_{1-c} respectively. Similarly, $\tilde{\boldsymbol{x}}$ is divided into $\tilde{\boldsymbol{x}}_{I_c}$ and $\tilde{\boldsymbol{x}}_{I_{1-c}}$.
8. *Alice* sends $\boldsymbol{y}_0 = \mathsf{Encode}(\tilde{\boldsymbol{r}}_0) \oplus (\boldsymbol{x}_{I_0})$, $\boldsymbol{z}_0 = \boldsymbol{m}_0 \oplus f_0(\tilde{\boldsymbol{r}}_0)$, $\boldsymbol{y}_1 = \mathsf{Encode}(\tilde{\boldsymbol{r}}_1) \oplus (\boldsymbol{x}_{I_1})$, and $\boldsymbol{z}_1 = \boldsymbol{m}_1 \oplus f_1(\tilde{\boldsymbol{r}}_1)$ to Bob, where $\tilde{\boldsymbol{r}}_0, \tilde{\boldsymbol{r}}_1$ are both random bit strings, Encode is the encoder of an error-correcting code, and f_0, f_1 are both hash functions.
9. *Bob* calculates $\boldsymbol{m}_c = \boldsymbol{z}_c \oplus f_c(\mathsf{Decode}(\boldsymbol{y}_c \oplus (\tilde{\boldsymbol{x}}_{I_c})))$, where Decode is the decoder of the error-correcting code.

Fig. 2 | Quantum Oblivious Transfer Protocol. Protocol flow for the one-out-of-two quantum oblivious transfer implementation used in this work. The steps include state preparation, measurement, commitment and opening checks, set partitioning, error-correction-based masking, and final message recovery by Bob for exactly one selected branch.

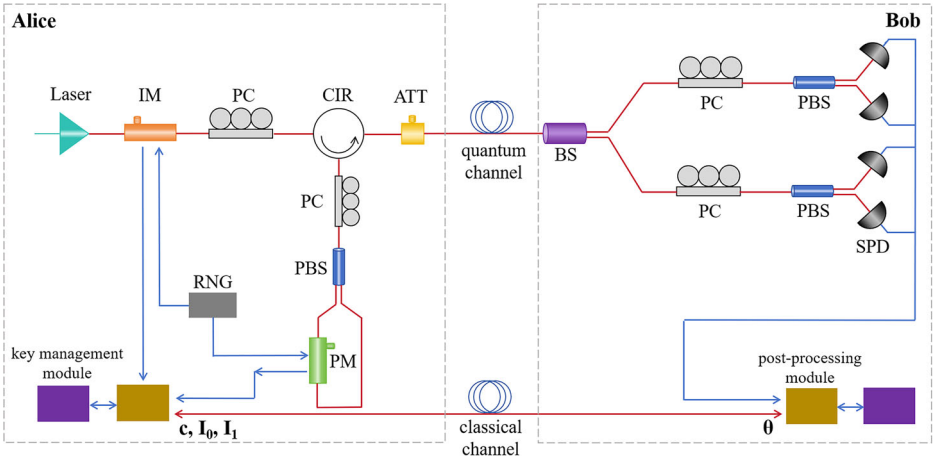


Fig. 3 | Schematic of experimental setup. Alice uses an intensity modulator with decoy intensity modulation. A circulator, a polarization controller, a polarization beam splitter, and a phase modulator constitute the polarization encoding part. The intensity modulator and the phase modulator are controlled by a random number generator. The light is then attenuated to a single-photon level with an attenuator and sent to Bob through fiber for detection. Bob uses a beam splitter to measure in different bases, and each basis is decoded by a polarization controller and a polarization beam splitter and detected by single-photon detectors. Post-processing and key management are performed by the corresponding modules, and an additional fiber channel is used for classical communication.

Table 1 | Dependence of the QOT code rate on optical attenuation

Optical attenuation (dB)	Code Rate (kbps)
-0.15	66.66
-1.97	66.11
-3.97	66.66
-5.96	66.66
-8.06	66.46
-9.69	66.66
-11.81	50.00
-13.44	36.66
-15.58	25.00
-17.88	17.77
-20.21	11.11
-22.18	5.00
-24.42	3.33
-26.69	0

The code rate is defined as the number of successfully transmitted qubits per second (kbps).

find that the cost for a malicious Bob to successfully bypass the bit commitment scheme and steal both of Alice's messages is exceedingly high, approximately 3.8×10^{12}. Here the cost is defined as T/p, where T represents the running time of Bob's attack algorithm, and p is the probability of a successful cheating attempt.

To provide an intuitive interpretation: with $\text{cost}_{\text{cheat}} \approx 3.8 \times 10^{12}$, if one run of the protocol takes one second, then Bob would need on average about 120,000 years to succeed in cheating once. This makes the cheating probability negligible in any realistic deployment.

Detailed calculations can be found in Methods 4.3.

Experimental implementation of QOT

We experimentally implement the above QOT protocol where both Alice and Bob are honest. We adopt the decoy-state BB84 protocol in our experiment. Alice uses a weak coherent laser source and the decoy state method[29–31] to prevent photon number splitting attacks[28,29]. The schematic implementation setup is illustrated in Fig. 3. An intensity modulator (IM) is used to implement the method with three different mean photon numbers: μ, ν, and 0, representing signal, decoy, and vacuum states, respectively. The mean photon number of each pulse is randomly modulated among μ, ν, and 0 with optimized probabilities p_1, p_2, and $1 - p_1 - p_2$, respectively. After polarization by a polarization controller (PC), the light enters a circulator for encoding.

Polarization encoding is achieved using a PC, a polarization beam splitter (PBS), and a phase modulator (PM). The PC adjusts the angle between the input light and the optical axis of the PBS to 45°. After passing through the PBS, the two polarization components travel clockwise and counterclockwise through the same loop, are combined at the PBS, and then exit the circulator. By precisely controlling the position of the PM in the loop and the timing of the modulation signal, four BB84 polarization states are modulated. The IM and PM are both controlled by a random number generator (RNG) to ensure randomness. The signals are then attenuated to a single-photon level with an attenuator and transmitted to Bob via fiber.

On the receiving side, Bob splits the light into two paths with a beam splitter (BS) for measurement on the Z basis or X basis. Each basis is adjusted by a PC to align with the sender's basis, followed by a PBS and two single-photon detectors (SPDs) to perform projective measurements. The post-processing modules receive modulation information from the IM and PM, and detection results from the SPDs for data processing. The key management modules are used for the storage, management, input, and output of authentication keys. Two post-processing modules communicate with each other through a classical fiber channel.

We test the functionality of our QOT protocol using the above physical device. The code rates of our QOT protocol for different optical decays are shown in Table 1. Increasing optical attenuation raises the channel loss, which reduces the number of detected photons and thus lowers the code rate.

The security of the protocol is not compromised by a higher bit error rate. Security relies on the bit commitment and the testing of a random subset: if the error rate becomes too high, the protocol will abort during the test phase (Step 4). Otherwise, the remaining errors are handled by the error correcting code (BCH code) and privacy amplification, which guarantee that even with a moderate error rate, Bob's information about the non selected message remains negligible. Therefore, while attenuation affects the practicality (throughput and maximum distance), it does not weaken the information theoretic security guarantees.

Application to the PSI problem

Upon completing the QOT protocol, we apply it to address a PSI problem in practice: identifying accounts blacklisted by one bank that also appear in another bank's list of accounts, without disclosing any other clients' information. PSI can be implemented by using an Oblivious Pseudorandom Function (OPRF) protocol, which can be further realized by our QOT protocol.

A simplified PSI solution evolving two parties Alice and Bob using OPRF can be achieved as follows. Denote Alice's private set as $\mathcal{X}$ and Bob's private set as $\mathcal{Y}$. Alice first generates a secret key $\mathbf{k}$. Alice encrypts $\mathcal{X}$ as

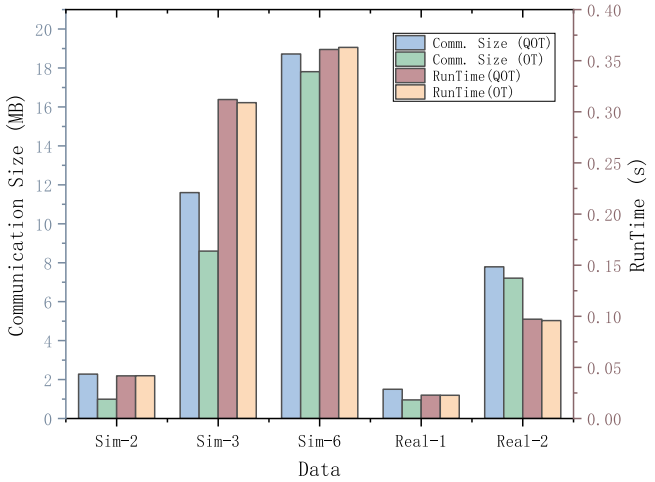


Fig. 4 | Comparison of communication cost and runtime between quantum oblivious transfer based and classical oblivious transfer based private set intersection experiments. Sim-1, Sim-2, and Sim-3 correspond to simulation datasets with different data sizes. Real-1 and Real-2 correspond to real-world datasets. From left to right, the number of queries in each private set intersection experiment is 10^3, 10^3, 10^5, 10^4, and 10^5, respectively, and the number of elements in the private set in each experiment is 10^4, 10^5, 10^5, 10^4, and 10^4, respectively.

$\mathcal{A} = \{f(\mathbf{k}, x) : x \in \mathcal{X}\}$, where f is a pseudorandom function, hardly distinguishable from a genuinely random function. Then Alice and Bob execute OPRF protocol, enabling Bob, as a receiver, to evaluate $\mathcal{B} = \{f(\mathbf{k}, y) : y \in \mathcal{Y}\}$, without learning Alice's secret key $\mathbf{k}$. At the same time, Alice, as a sender, remains oblivious to Bob's input y and the associated output $f(\mathbf{k}, y)$. Finally, Alice sends $\mathcal{A}$ to Bob, and Bob computes $\mathcal{A} \cap \mathcal{B}$ to figure out the intersection of private sets $\mathcal{X}$ and $\mathcal{Y}$.

In this work, we integrate our QOT protocol into the optimized OPRF/PSI presented in ref. 32. The details are given in Section 4.5. We conducted the PSI experiment on both simulated and real-world data. For real-world data, one bank provides blacklist accounts potentially involving telecommunication fraud among multiple institutions, while the other bank, acting as an inquier, supplies active account data for the matching task. We want to find the intersection of two lists without revealing any further information.

Our experiments demonstrate scalability for PSI with set sizes up to 10^5 elements. The communication cost and runtime grow nearly linearly with the number of elements, as shown in Fig. 4. For instance, with 10^5 elements per party, the total communication is about 20 MB and the runtime is below 0.4 s on a 1Gbps network. These numbers are already practical for many real world applications (e.g., fraud detection in banking). Further scalability can be achieved by combining our QOT with classical OT extension techniques, which allow generating a large number of OTs from a small number of base OTs.

We apply our QOT protocol to solve this PSI problem, successfully identifying highly suspicious telecommunications fraud bank accounts. To evaluate the performance of our protocol, we measure the communication cost and runtime during the PSI experiment, with the results illustrated in Fig. 4. To benchmark the performance of our QOT protocol, we also compare the communication cost and runtime of the same PSI experiment based on classical OT in Fig. 4. Our results indicate that QOT incurs only slightly higher communication overhead compared to classical OT, while their runtime remains nearly the same. In exchange, our QOT protocol provides quantum security that classical OT cannot achieve. Our experiment demonstrates the feasibility of the above QOT protocol.

Discussion

In this paper, we present an experimental implementation of a protocol that is secure against any quantum attack. Unlike QOT protocols in the noisy storage model, our protocol does not rely on assumptions about noisy or limited quantum memory on the part of the adversary. Instead of restricting the capability of the adversary, our protocol employs a bit commitment scheme to guarantee security against delayed measurement attacks. The security of our QOT protocol is based on the well-founded security of bit commitment. Our protocol is secure against both classical and quantum attacks, within polynomial time. The experimental demonstration uses an optical system adapted from a decoy state QKD system, ensuring security against photon splitting attacks in practical single-photon sources.

While it is well known that secure MPC can be realized through OT[6], no previous experiments have utilized quantum oblivious transfer (QOT) to solve real-world problems. In this paper, we are the first to apply our QOT protocol to address a practically significant PSI problem in finance. Specifically, we demonstrate how two banks, one with a blacklist and the other with a list of active accounts, can compute the intersection of their private data sets without revealing any further information to each other. Our framework, when applied to other problem instances, enables secure MPC applications beyond PSI, including privacy-preserving machine learning[4] and anonymous voting systems[33].

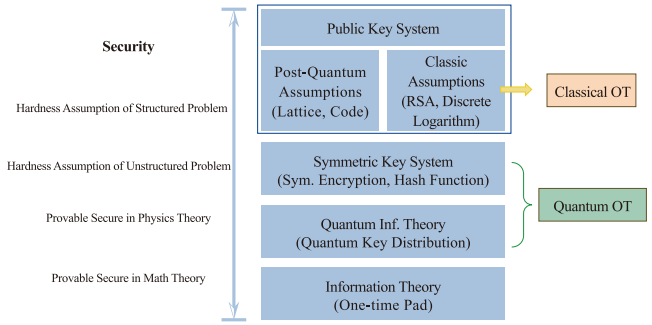


Fig. 5 | Hierarchy of cryptographic assumptions. The assumptions become weaker from top to bottom, and the corresponding cryptographic protocols are considered increasingly reliable.

Methods

Cryptographic assumption hierarchy

Cryptographic primitives can be broadly categorized into three types: information-theoretic, symmetric-key, and public-key systems. The one-time pad is a prime example of information-theoretic security, providing unconditional security that can be rigorously proven mathematically. Symmetric-key systems, including symmetric encryption and hash functions, are based on complex, unstructured problems. In contrast, public-key systems depend on structured hard problems with algebraic representations, such as RSA. While the security of both symmetric-key and public-key systems is based on the presumed hardness of their underlying problems, symmetric-key systems are considered more foundational due to their reliance on weaker assumptions.

A further distinction exists between traditional cryptographic assumptions, like RSA and discrete logarithms, which are vulnerable to quantum attacks[34], and other assumptions, such as lattice-based and code-based cryptography, which are not currently known to be susceptible to quantum algorithms[35]. Quantum information theory relies on established principles of quantum physics, which, if disproven, would indicate a discovery of new physics. QKD is typically classified as information-theoretic, representing the highest level of security. OT usually employs public-key cryptography for implementation. However, as noted in the literature[12,24], OT protocols can also be constructed using only quantum information and symmetric-key encryption. This approach offers more reliable assurance without relying on strong computational assumptions.

As illustrated in Fig. 5, our QOT protocol is based on weaker cryptographic assumptions compared to classical OT (or even post-QOT). Specifically, our protocol requires only quantum-secure bit commitment schemes, which depend on the existence of (quantumly hard) one-way

Table 2 | Security assumptions for OT protocols

Protocol	Assumption	Type
7,8	Discrete Logarithm / Factoring	Structured
41	LWE	Structured
13	Quantum + Noisy Memory	Physical Limitation
This work	**Quantum + Hash Functions**	**Unstructured**

permutations or collision-resistant hash functions-both categorized as unstructured symmetric mathematical problems. In contrast, post-QOT and public-key encryption typically rely on the conjectured hardness of lattice-based or code-based problems. This suggests that, while it is conceivable (though less likely) that the hardness of lattice and code problems could be disproven in the future, the existence of one-way permutations and collision-resistant hash functions is more likely to hold true. Moreover, the existence of (quantumly hard) one-way functions represents the minimal assumption necessary for meaningful cryptography beyond secure communication in the quantum realm.

To clarify the relative strength of assumptions, we provide a detailed comparison in Table 2. The security of our QOT protocol relies on the existence of quantum secure bit commitment, which can be built from collision resistant hash functions which belongs to the class of "unstructured" symmetric key problems. Our protocol therefore operates under a strictly weaker set of assumptions: it requires only the minimal assumption for meaningful cryptography in the quantum setting. The table below summarizes this comparison:

By basing security on cryptographic hash functions, our protocol achieves quantum security while relying on the most foundational and long standing cryptographic primitives. This makes it more robust against future advances that might break structured problems or noisy quantum memory assumption.

QOT protocol

Now we explain our QOT protocol step by step. Alice, as the sender, has three optical sources, one of which is for preparing signal states while the other two preparing decoy states. At each round i, Alice chooses two bits $x_i \in \{0,1\}$ and $\theta_i \in \{0,1\}$ independently at random. By randomly using one of her optical sources, Alice prepares a polarization-encoded single-photon state $|x_i\rangle_{\theta_i}$ and then sends her encoded state to Bob. At the receiver's side, Bob randomly chooses a bit $\widetilde{\theta}_i \in \{0,1\}$ and performs a projective measurement on the i-th received qubit in the basis $\{|0\rangle_{\widetilde{\theta}_i}, |1\rangle_{\widetilde{\theta}_i}\}$, where $|0\rangle_0 := |0\rangle$, $|1\rangle_0 := |1\rangle$, $|0\rangle_1 := |+\rangle$, and $|1\rangle_1 := |-\rangle$.

Due to imperfections in the photon source and dark counts, Bob may fail to detect any photons in certain rounds. Bob reports to Alice that rounds had no photon clicks at his detector. Alice then checks whether the ratio of Bob's reported rounds of photon loss falls within a reasonable range for each decoy state source, based on the estimated parameters of their quantum devices. If Bob's reported ratio is outside this reasonable range, Alice concludes that Bob must be cheating and abort the protocol. Otherwise, Alice proceeds as follows.

The set of rounds where Alice sent signals while Bob reported photon clicks is denoted by $\mathcal{M}$ and the cardinality of $\mathcal{M}$ is $|\mathcal{M}| = n$. To ensure that Bob measures all these n qubits instead of storing some qubits in his quantum memory for measurements after Alice reveals her bases $\boldsymbol{\theta} := \{\theta_i | i \in \mathcal{M}\}$, Bob must commit to all his values of $\widetilde{\boldsymbol{\theta}} := \{\widetilde{\theta}_i | i \in \mathcal{M}\}$ and $\widetilde{\boldsymbol{x}} := \{\widetilde{x}_i | i \in \mathcal{M}\}$. This step is the key difference between our QOT protocol from the one in the noisy storage model[16].

In our protocol, after Bob measures the received qubits, he must commit to all his measurement bases and outcomes before Alice reveals her bases. This is done via a collision resistant hash function: for each round i, Bob computes $c_i = h(\widetilde{x}_i, \widetilde{\theta}_i, r_i)$ where h is a hash function (e.g., SHA 256) and r_i is a random string, and sends c_i to Alice. Later, when Alice requests to open a subset of these commitments (the test set T), Bob reveals $(\widetilde{x}_i, \widetilde{\theta}_i, r_i)$ for $i \in T$. Alice verifies that the opened values match the hash and that the bit error rate on matching bases is below a threshold.

This bit commitment step is crucial for security against delayed measurement attacks: if Bob could store qubits indefinitely, he could wait for Alice to announce her bases θ and then measure each qubit in the correct basis, thereby learning both x_i values (for both bases) and breaking the OT. By forcing Bob to commit to his measurements before θ is revealed, we ensure that any deviation (e.g., storing qubits instead of measuring) will be detected with high probability during the test phase. The commitment scheme binds Bob to his measurement choices, so he cannot adaptively change them after learning θ.

In our experimental implementation, we use a practical bit commitment scheme based on the cryptographic hash function SHA-256. Concretely, to commit to a bit pair $(\widetilde{x}_i, \widetilde{\theta}_i)$, Bob chooses a random 256 bit string r_i and computes the commitment as $c_i = \mathrm{SHA}-256(r_i \| \widetilde{x}_i \| \widetilde{\theta}_i)$, where $\|$ denotes concatenation. When opening, Bob reveals $(r_i, \widetilde{x}_i, \widetilde{\theta}_i)$, and Alice recomputes the hash to verify.

To verify Bob's commitment to his measurement bases, Alice randomly chooses a subset $T \subset \mathcal{M}$ of size $|T| = \alpha n$, where $0 < \alpha < 1$, as a test set and reveals the committed values c_i for $i \in T$. To open a commitment c_i, Alice asks Bob to send her the values of $\widetilde{x}_i$, $\widetilde{\theta}_i$ and r_i. Due to the collision resistance property of h, Alice can easily find out whether Bob changed his committed value. Then, Alice checks whether $\widetilde{x}_j = x_j$ holds true when $\widetilde{\theta}_j = \theta_j$ for $j \in T$ with a ratio at least $\beta \in (1/2, 1)$. If it holds, then Alice passes this verification and continues; otherwise, she aborts. After that, both Alice and Bob discard the test qubits in the set T, and Alice sends all her bases $\{\theta_i | i \in \mathcal{M} \backslash T\}$ to Bob.

When $\theta_i = \widetilde{\theta}_i$, the bits x_i and $\widetilde{x}_i$ match with a high rate, but appear independently random otherwise. While Bob knows which ones correspond to $\theta_i = \widetilde{\theta}_i$, Alice remains unaware of this information. Bob randomly chooses a choice bit $c \in \{0,1\}$ and divides the set $\mathcal{M} \backslash T$ into two subsets: a good set $I_c := \{i \in \mathcal{M} \backslash T | \theta_i = \widetilde{\theta}_i\}$ and a bad set $I_{1-c} := \{i \in \mathcal{M} \backslash T | \theta_i \neq \widetilde{\theta}_i\}$. The bit strings $\boldsymbol{x}$ is then divided into two $\boldsymbol{x}_{I_c}$ and $\boldsymbol{x}_{I_{1-c}}$ on I_c and I_{1-c} respectively. Similarly, $\widetilde{\boldsymbol{x}}$ is divided into $\widetilde{\boldsymbol{x}}_{I_c}$ and $\widetilde{\boldsymbol{x}}_{I_{1-c}}$. For simplicity, we suppose $|I_c| = |I_{1-c}|$. In the case where I_c and I_{1-c} have different lengths, we truncate the longer one to the length of the other. Bob sends both sets I_0 and I_1 to Alice, without informing her which one corresponds to the good one.

Due to transmission errors as well as measurement errors, even for $i \in I_c$, there is still some chance that $x_i \neq \widetilde{x}_i$. To reduce such bit-flip errors in practice, Alice and Bob employ a classical error correction code to construct noiseless shared random secret keys. For example, they can use Bose-Chaudhuri-Hocquenghem (BCH) codes[36] that are characterized by their ability to correct multiple random errors. Alice generates random secret keys $\widetilde{\boldsymbol{r}}_0$ and $\widetilde{\boldsymbol{r}}_1$ of the same length and uses an error correction code to encode them. Then she sends to Bob $\boldsymbol{y}_0 := \mathrm{Encode}(\widetilde{\boldsymbol{r}}_0) \oplus \boldsymbol{x}_{I_0}$ and $\boldsymbol{y}_1 := \mathrm{Encode}(\widetilde{\boldsymbol{r}}_1) \oplus \boldsymbol{x}_{I_1}$, where Encode denotes the encoding function of the error correction code and $\oplus$ denotes addition modulo two. After receiving $\boldsymbol{y}_0$ and $\boldsymbol{y}_1$, Bob uses his bit strings $\widetilde{\boldsymbol{x}}_{I_c}$ to calculate $\mathrm{Decode}\left(\boldsymbol{y}_c \oplus \widetilde{\boldsymbol{x}}_{I_c}\right)$, where Decode denotes the decoding function of the error correction code. When the number of bit flip errors $\sum_{i \in I_c} |x_i \oplus \widetilde{x}_i|$ is below a certain error threshold, Bob can perfectly recover $\widetilde{\boldsymbol{r}}_c = \mathrm{Decode}(\mathrm{Encode}(\widetilde{\boldsymbol{r}}_c) \oplus \boldsymbol{x}_{I_c} \oplus \widetilde{\boldsymbol{x}}_{I_c})$. On the other hand, as $\boldsymbol{x}_{1-c}$ and $\widetilde{\boldsymbol{x}}_{1-c}$ are independently random, Bob can rarely learn information about $\widetilde{\boldsymbol{r}}_{1-c}$.

Up to now, Alice has obliviously transferred random keys $\widetilde{\boldsymbol{r}}_0$ and $\widetilde{\boldsymbol{r}}_1$ to Bob, with Bob only receiving $\widetilde{\boldsymbol{r}}_c$. Alice can now use these obliviously transferred random secret keys to accomplish the OT protocol. However, the probability that Bob can learn information about $\widetilde{\boldsymbol{r}}_{1-c}$ is nonzero. To minimize the information Bob can retrieve about $\widetilde{\boldsymbol{r}}_{1-c}$, we use privacy amplification, ensuring that Bob ultimately learns nothing about the secret key. By using two universal hash functions f_0 and f_1 agreed upon by both Alice and Bob, they obtain bit strings $f_0(\widetilde{\boldsymbol{r}}_0)$ and $f_1(\widetilde{\boldsymbol{r}}_1)$ of shorter length λ from the original random keys $\widetilde{\boldsymbol{r}}_0$ and $\widetilde{\boldsymbol{r}}_1$. Suppose Alice has two pieces of messages $\boldsymbol{m}_0$ and $\boldsymbol{m}_1$ of length λ. Using the secret keys $f_0(\widetilde{\boldsymbol{r}}_0)$ and $f_1(\widetilde{\boldsymbol{r}}_1)$ as

one-time pads, Alice sends Bob her encrypted messages $\boldsymbol{z}_0 := \boldsymbol{m}_0 \oplus f_0(\widetilde{\boldsymbol{r}}_0)$ and $\boldsymbol{z}_1 := \boldsymbol{m}_1 \oplus f_1(\widetilde{\boldsymbol{r}}_1)$. Hence, Bob uses $f_c(\widetilde{\boldsymbol{r}}_c)$ to recover the message $\boldsymbol{m}_c = \boldsymbol{z}_c \oplus f_c(\widetilde{\boldsymbol{r}}_c)$ correctly. At the same time, since Alice does not know the value of c, she is oblivious to which of $\boldsymbol{m}_0$ and $\boldsymbol{m}_1$ has been successfully sent to Bob throughout the entire protocol.

Correctness and security analysis

Let Bin(n, p) denote a binomial distribution. For a random variable $X \sim$ Bin(n, p), define $B_{n,p,\geq m} = \Pr[X \geq m] = \sum_{i=m}^{n} \binom{n}{i} p^i (1-p)^{n-i}$ and $B_{n,p,\leq m} = \Pr[X \leq m] = \sum_{i=0}^{m} \binom{n}{i} p^i (1-p)^{n-i}$

Assuming independent and identical noise in each round, let the overall bit flip error rate be p_e. We take p_e = 0.01 in the analysis. The correctness of our protocol relies on two parts. First, Bob must pass the check in Steps 3-5. Second, the number of matching bits in set I_c, i.e. $|\{i \in I_c | \widetilde{x}_i = x_i\}|$, must be at least $N - t$ to ensure that the message can be decoded in Step 9. Therefore, the failure probability is $p_{\text{failed}} \leq p_{\text{fpass}} + p_{\text{fcorrect}}$,

Table 3 | The parameter choices used in our experimental implementation

physical meaning of parameters	values
length of message	$\lambda = 256$
number of single photon states received by Bob	$n = 2044$
ratio of test bits	$\alpha = 1/2$
minimum acceptable ratio of bit matching	$\beta = 0.95$
error correction code length	$N = 511$
maximum number of correctable bit flip errors	$t = 30$

where

$$p_{\text{fpass}} = \sum_{i=0}^{\beta\alpha n} \binom{\alpha n}{i} p_e^{\alpha n - i} (1 - p_e)^i. \tag{1}$$

$$p_{\text{fcorrect}} = 1 - \frac{1}{2^{n-\alpha n}} \sum_{i=N+1}^{n-\alpha n} \left(\binom{n-\alpha n}{i} \cdot \left(\sum_{j=N-t}^{N} \binom{N}{j} p_e^{N-j} (1-p_e)^j \right) \right) - \frac{1}{2^{n-\alpha n}} \sum_{i=0}^{N} \left(\binom{n-\alpha n}{i} \cdot \left(\sum_{j=0}^{i} \binom{i}{j} p_e^{i-j} (1-p_e)^j \cdot \frac{1}{2^{N-i}} \sum_{k=N-t-j}^{N-i} \binom{N-i}{k} \right) \right). \tag{2}$$

Using the parameters listed in Table 3, we find that the probability is at most 0.177%, which is quite close to zero.

Next, let us consider the security of our protocol against a malicious Bob. Bob can attempt to steal information from Alice by delaying his measurement in step 2, until Alice reveals the correct basis in step 6. However, he must first bypass the checks in steps 3–5. Assuming $p_e = 0$, which is favorable for Bob, he can delay the measurement of s_1 qubits (where $s_1 \leq \alpha n$) in step 2 until Alice reveals her bases and guess the measurement outcomes of s_2 qubits in the set I_{1-c}. The probability that he bypasses the first check is

$$p_{\text{bypass},s_1} = \frac{1}{\binom{n}{\alpha n}} \cdot \sum_{i=0}^{s_1} \binom{s_1}{i} \binom{n-s_1}{\alpha n - i} \Pr[\text{check passed}], \tag{3}$$

where

$$\Pr[\text{check passed}] = \begin{cases} 1 & \text{if } \alpha n - \mathrm{i} \geq \beta\alpha n, \\ \sum_{\mathrm{j}=\lceil \mathrm{i}-(1-\beta)\alpha n \rceil}^{\mathrm{i}} \binom{\mathrm{i}}{\mathrm{j}} \frac{1}{2^{\mathrm{j}}} \cdot \frac{1}{2^{\mathrm{i}-\mathrm{j}}} & \text{otherwise}. \end{cases} \tag{4}$$

Protocol OPRF

Input of $\mathcal{R}$: m selection strings $r = (r_1, \ldots, r_m)$, $r_i \in \{0,1\}^*$.
Parameters:

- A (κ, ϵ)-PRC family $\mathcal{C}$ with output length $k = k(\kappa)$.
- A κ-Hamming correlation-robust $H : [m] \times \{0,1\}^k \to \{0,1\}^v$.
- An ideal OT_m^k primitive.

Protocol:

1. $\mathcal{S}$ chooses a random $C \leftarrow \mathcal{C}$ and sends it to $\mathcal{R}$.
2. $\mathcal{S}$ chooses $s \leftarrow \{0,1\}^k$ at random. Let s_i denote the i-th bit of s.
3. $\mathcal{R}$ forms $m \times k$ matrices T_0, T_1 in the following way:
   - For $j \in [m]$, choose $t_{0,j} \leftarrow \{0,1\}^k$ and set $t_{1,j} = C(r_j) \oplus t_{0,j}$.

   Let t_i^0, t_i^1 denote the i-th column of matrices T_0, T_1 respectively.
4. $\mathcal{S}$ and $\mathcal{R}$ interact with OT_m^k in the following way:
   - $\mathcal{S}$ acts as receiver with input $\{s_i\}_{i \in [k]}$.
   - $\mathcal{R}$ acts as sender with input $\{t_i^0, t_i^1\}_{i \in [k]}$.
   - $\mathcal{S}$ receives output $\{q^i\}_{i \in [k]}$.

   $\mathcal{S}$ forms $m \times k$ matrix Q such that the i-th column of Q is the vector q^i. (Note $q^i = t_{s_i}^i$.) Let q_j denote the j-th row of Q. Note, $q_j = ((t_{0,j} \oplus t_{1,j}) \cdot s) \oplus t_{0,j}$. Simplifying, $q_j = t_{0,j} \oplus (C(r_j) \cdot s)$.
5. For $j \in [m]$, $\mathcal{S}$ outputs the PRF seed $((C, s), (j, q_j))$.
6. For $j \in [m]$, $\mathcal{R}$ outputs relaxed PRF output $(C, j, t_{0,j})$.

Fig. 6 | Oblivious PRF Protocol. Detailed algorithmic description of the batched related-key oblivious pseudorandom function construction used as a building block for the private set intersection implementation.

Protocol PSI

Parameters: Alice has input X; Bob has input Y, with $|X| = |Y| = n$. s is an upper bound on the stash size for Cuckoo hashing.

1. Bob specifies random hash functions $h_1, h_2, h_3 : \{0,1\}^* \to [1.2n]$ and tells them to Alice.
2. Bob assigns his items Y into $1.2n$ bins using Cuckoo hashing. Let Bob keep track of $z(y)$ for each y so that: if $z(y) = \perp$ then y is in the stash; otherwise y is in bin $h_{z(y)}(y)$. Arrange the items in the stash in an arbitrary order.

   Bob selects OPRF inputs as follows: for $i \in [1.2n]$, if bin i is empty, then set r_i to a dummy value; otherwise if y is in bin i then set $r_i = y \| z(y)$. For $i \in [s]$, if position i in the stash is y, then set $r_i = y$; otherwise set r_i to a dummy value.
3. The parties invoke $1.2n + s$ OPRF instances, with Bob the receiver with inputs $(r_1, \ldots, r_{1.2n+s})$. Alice receives $(k_1, \ldots, k_{1.2n+s})$ and Bob receives $F(k_i, r_i)$ for all i.
4. Alice computes $H_i = \{F(k_{h_i(x)}, x \| i) \mid x \in X\}$ for $i \in \{1,2,3\}$, and $S_j = \{F(k_{1.2n+j}, x) \mid x \in X\}$ for $j \in \{1, \ldots, s\}$, and sends a permutation of each set to Bob.
5. Bob initializes an empty set $\mathcal{O}$ and does the following for $y \in Y$: If $z(y) = \perp$ and y is at position j in the stash and $F(k_{1.2n+j}, y) \in S_j$, then Bob adds y to $\mathcal{O}$. If $z(y) \neq \perp$ and $F(k_{h_{z(y)}(y)}, y \| z(y)) \in H_{z(y)}$ then Bob adds y to $\mathcal{O}$.
6. Bob sends $\mathcal{O}$ to Alice and both parties output $\mathcal{O}$.

Fig. 7 | Private Set Intersection Protocol. Detailed algorithmic description of the private set intersection protocol instantiated from batched oblivious pseudorandom function calls, including data placement, protocol execution, and output reconstruction.

Once Bob bypasses the first check, he must obtain the correct values for more than $2(N-t)$ qubits to acquire both messages. He can get s_1 bits from the associated qubits delayed for measurements and another s_2 bits from guessing. The probability of this happening is

$$p_{\text{cheat},s_1,s_2} = \frac{1}{2^{(1-\alpha)n}} \sum_{i=2(N-t)-s_1-s_2}^{(1-\alpha)n} \binom{(1-\alpha)n}{i}. \tag{5}$$

By defining the cheating cost as the expected time for an attacker to successfully deceive once, Bob's final cheating cost is

$$\text{cost}_{\text{cheat}} = \min_{0 \le s_1 \le \alpha, 0 \le s_2 \le 2(N-t)-s_1} \frac{2^{s_2}}{p_{\text{bypass},s_1} \cdot p_{\text{cheat},s_1,s_2}} \ge 3.8 \times 10^{12}. \tag{6}$$

To offer an intuitive understanding: given that $\text{cost}_{\text{cheat}} \approx 3.8 \times 10^{12}$, if each execution of the protocol lasts one second, Bob would require roughly 120,000 years on average to successfully cheat just once. Thus, in any practical application, the probability of cheating becomes essentially negligible.

Potential Optimization

Our QOT protocol, as well as OT protocol, has some potential optimization methods to reduce the computation and communication cost.

OT extension enables us to extend the number of OTs performed using a small set of base OTs. Ishai et al.[37] introduced a framework for OT extension within the Random Oracle Model (ROM), which only makes use of cheap symmetric key primitives. Following studies have built upon the Ishai et al.[37] framework, subsequent works enhancing it to increase security level with better efficiency[38,39].

Beaver triples[40] are pre-shared correlated randomness and are widely used in secure MPC. Beaver triple can be pre-generated by OT in the offline phase. In the online phase, parties can execute MPC protocol by beaver triples without executing OT.

PSI from oblivious pseudorandom function

The Batched, Related-Key Oblivious Pseudorandom Function (BaRK-OPRF) protocol[32] is an efficient implementation of OPRF. This is accomplished through a batchable approach, where multiple evaluations can be handled efficiently in a single protocol execution. Based on the BaRK-OPRF protocol,[32] presents an improved PSI protocol. The detailed algorithm is listed in Fig. 6 and Fig. 7.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to confidentiality agreements with participating financial institutions and privacy restrictions on customer-related data, but are available from the corresponding author on reasonable request. No publicly available sequence data were used in this study; therefore accession numbers are not applicable.

Code availability

The code that supports the findings of this study is available from the corresponding author upon reasonable request.

Materials availability

The materials that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Kai-Yi Zhang, Ya-Dong Wu and Yu Yu designed research; An-Jing Huang, Kun Tu, Ming-Han Li, Chi Zhang and Wei Qi performed the experiment. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Pending patents

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Additional information

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